INJECTIVE ON POINTS WITH SURJECTIVE STALK MAPS IS A MONOMORPHISM

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Editorial note. Counted span is the author's own proof environment, source lines 4840-4853, following the statement at 4827-4838.

Lemma 0.1. *Let $j : X \to Y$ be a morphism of schemes. If*

(1) *j is injective on points, and*
(2) *for any $x \in X$ the ring map $j^\sharp_x : \mathcal{O}_{Y,j(x)} \to \mathcal{O}_{X,x}$ is surjective,*

then j is a monomorphism.

Proof. Let $a, b : Z \to X$ be two morphisms of schemes such that $j \circ a = j \circ b$. Then (1) implies $a = b$ as underlying maps of topological spaces. For any $z \in Z$ we have $a^\sharp_z \circ j^\sharp_{a(z)} = b^\sharp_z \circ j^\sharp_{b(z)}$ as maps $\mathcal{O}_{Y,j(a(z))} \to \mathcal{O}_{Z,z}$. The surjectivity of the maps $j^\sharp_x$ forces $a^\sharp_z = b^\sharp_z$, $\forall z \in Z$. This implies that $a^\sharp = b^\sharp$. Hence we conclude $a = b$ as morphisms of schemes as desired. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.